Homology of Fibered Spaces

Ezra Guerrero 02/11/2026

Goal

- Use spectral sequences to understand the (co)homology of fibered spaces.
- Apply this theory to various applications, recovering important results.

Outline

① Review of Spectral Sequences

② Properties of Serre Fibrations

③ Applications

①

Filtrations

Spectral sequences show up in contexts when we have a filtration on our algebraic structure.

Let A be an abelian group and $d: A \to A$ a homomorphism such that $d^2 = 0$. Such a pair (A, d) is called a differential group. The family $(A^p)_{p \in \mathbb{Z}}$ of subgroups $A_p \subseteq A$ is an ascending filtration if the following three conditions hold:

1) $\bigcup_{p\in\mathbb{Z}} A^p = A$ 2) $A^p \subset A^{p+1}\ \forall p\in\mathbb{Z}$ 3) $d(A^p)\subseteq A^p$

Example

Let $A = \Phi_0 \oplus \Phi_1 \oplus \Phi_2$, $A^{-p} = 0$ for $p>0$, $A^0 = \Phi_0$, $A^1 = \Phi_0 \oplus \Phi_1$, and $A^p = A$ for $p \geq 2$. If this is an ascending filtration, then there are homomorphisms

$$f : \Phi_1 \to \Phi_0,\quad g : \Phi_2 \to \Phi_1,\quad h : \Phi_2 \to \Phi_0$$

and endomorphisms $d_j : \Phi_j \to \Phi_j$ such that

$$d = \begin{pmatrix} d_0 & f & h \\ 0 & -d_1 & g \\ 0 & 0 & d_2 \end{pmatrix}.$$

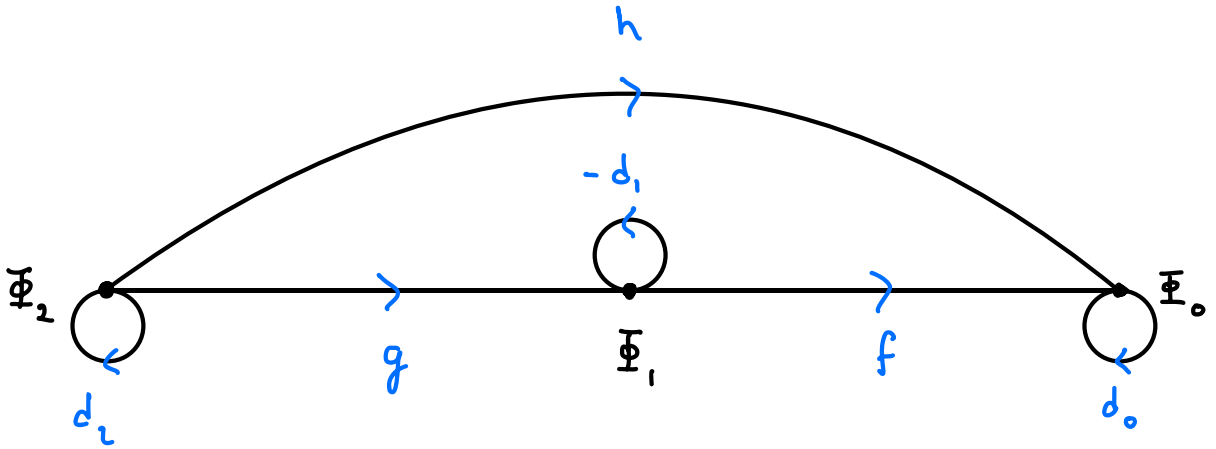
The requirement $d^2 = 0$, then reads

$$d_j^2 = 0; \tag{1}$$

$$d_0 f - f d_1 = 0; \tag{2}$$

$$-d_1 g + g d_2 = 0; \tag{3}$$

$$d_0 h + f g + h d_2 = 0. \tag{4}$$



Equation (1) gives that (E_j, d_j) is a differential group. Then, (2) and (3) imply that f, g induce maps on homology

$$f_* : H(E_1) \to H(E_0), \quad g_* : H(E_2) \to H(E_1).$$

Lastly, (4) says that $f_* g_* = 0$, giving a chain complex structure on homology.

This is exactly the iterated homological structure we will get from spectral sequences.

The Spectral Sequence

We want to look at chain complexes or, equivalently, graded differential groups. In this context, A is a direct sum

$$A = \bigoplus_{n \in \mathbb{Z}} A_n,$$

d has degree -1, so $d(A_n) \subseteq A_{n-1}$, and our filtration is compatible with the grading.

We write

$$A^{p,q} = A_{p+q} \cap A^p.$$

We are now ready to define the spectral sequence of A:

For $r \in \mathbb{N}$, let

$$C_r^{p,q} = \left\{ x \in A^p \mid dx \in A^{p-r} \right\} \cap A^{p,q}$$

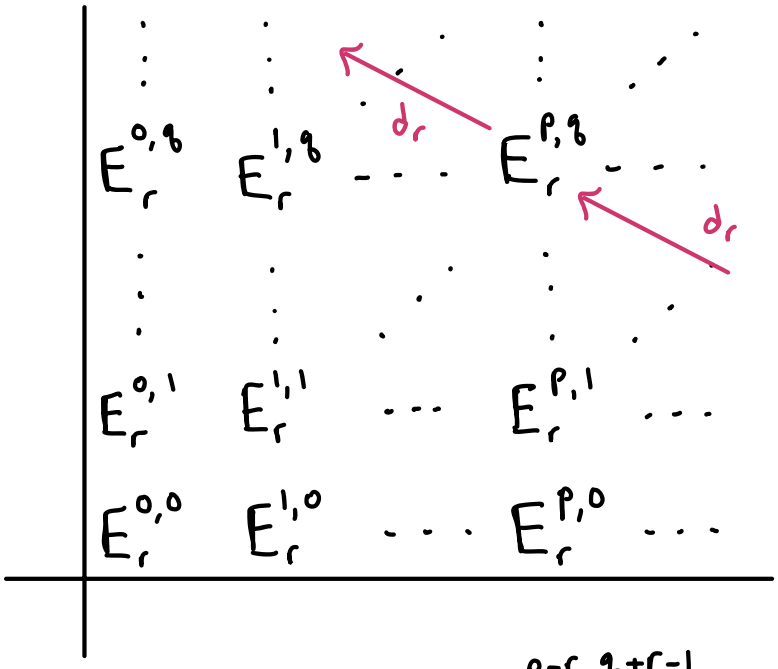
and

$$B_r^{p,q} = d(A^{p+r}) \cap A^{p,q}.$$

Moreover, let $C_\infty^{p,q} = \{x \in A^{p,q} \mid dx = 0\}$ and $B_\infty^{p,q} = d(A) \cap A^{p,q}$.

We define

$$E_r^{p,q} = C_r^{p,q} / \left(C_{r-1}^{p-1,q+1} + B_{r-1}^{p,q} \right).$$



Observe that $d(C_r^{p,q}) = B_r^{p-r,q+r-1} \subseteq C_r^{p-r,q+r-1}$ and $d(C_{r-1}^{p-1,q+1} + B_{r-1}^{p,q}) \subseteq C_{r-1}^{p-r-1,q+r}$. Therefore, for each r, d descends to a map

$$d_r : E_r^{p,q} \longrightarrow E_r^{p-r,q+r-1}$$

We can check that

1) $d_r^2 = 0$

2) $H(E_r) = E_{r+1}$.

For our work to be much nicer, we make the following assumption:

(A) $$A^{n,0} = A_n, \qquad A^{p,q} = 0 \quad \text{if } p < 0.$$

Proposition

If assumption (A) holds, then

$$E_1^{p,q} = E_{r+1}^{p,q} = \cdots E_\infty^{p,q} \qquad \text{for } r > \max(p, q+1)$$

Proof

If $r > p$, then $E_r^{p-r,q+r-1} = 0$, so $\ker\left(E_r^{p,q} \to E_r^{p-r,q+r-1}\right) = E_r^{p,q}$

and if $r > q+1$, then $E_r^{p+r,q-r+1} = 0$, so $\operatorname{im}\left(E_r^{p+r,q-r+1} \to E_r^{p,q}\right) = 0$.

Hence, $E_{r+1}^{p,q} = E_r^{p,q}$. To see they equal $E_\infty^{p,q}$, note that $C_r^{p,q}$ and $B_r^{p,q}$ are independent of r for r large enough.

In some sense, E_∞ is the "limit" of the E_r. Moreover, E_∞ is related to $H(A)$. Filter $H(A)$ by the images D^p of $H(A^p)$ in $H(A)$. Then, we find that

$$E_\infty^p = D^p / D^{p-1}.$$

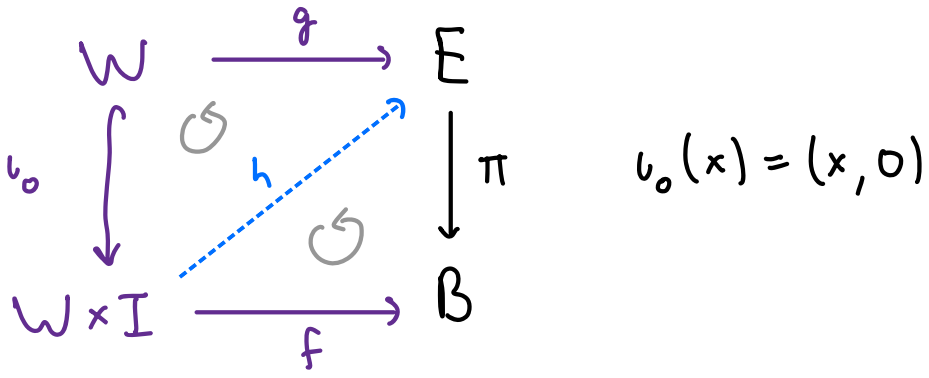
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Remark: Carrying out the analogous construction for a decreasing filtration, we get the cohomology spectral sequence.

Serre Fibrations

(Serre just called them fibrations, of course)

A Serre fibration is a map $\pi : E \to B$ that satisfies the homotopy lifting property for CW-complexes.



For any CW-complex W and maps $f : W \times I \to B$ and $g : W \to E$ making the above square commute, there exists some $h : W \times I \to E$ lifting f.

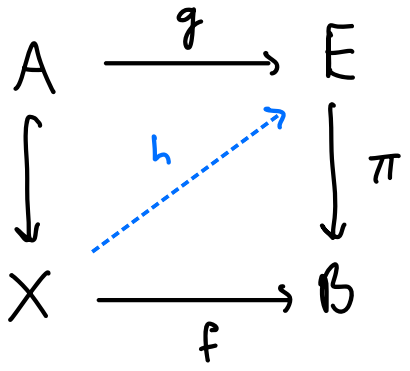
Examples

1) Fiber bundles are Serre fibrations.

2) Path spaces are Serre fibrations (coming soon)

Proposition

Let A, X be two contractible CW-complexes, with $A \subseteq X$. Let $\pi : E \to B$ be a Serre fibration and $f : X \to B$, $g : A \to E$ maps such that the following square commutes:



Then, there exists $h: X \to E$ extending g and lifting f.

This can be used to establish the homotopy long exact sequence:

$$\cdots \to \pi_n(F) \to \pi_n(E) \to \pi_n(B) \to \pi_{n-1}(F) \to \cdots$$

where $F = \pi^{-1}(b)$ is a fiber.

Cubes

We will use cubic (co)homology to study our spaces. It is not hard to show cubic homology coincides with singular homology. We use it to simplify arguments.

For a space X, an n-dimensional singular cube of X is a continuous map $u: I^n \to X$.

Such a cube is called degenerate if u does not depend on its last coordinate:

$$u(x_1, \ldots, x_{n-1}, x_n) = u(x_1, \ldots, x_{n-1}, x_n').$$

We define

$$Q_n(X) = \text{free abelian group on } n\text{-dim singular cubes}$$

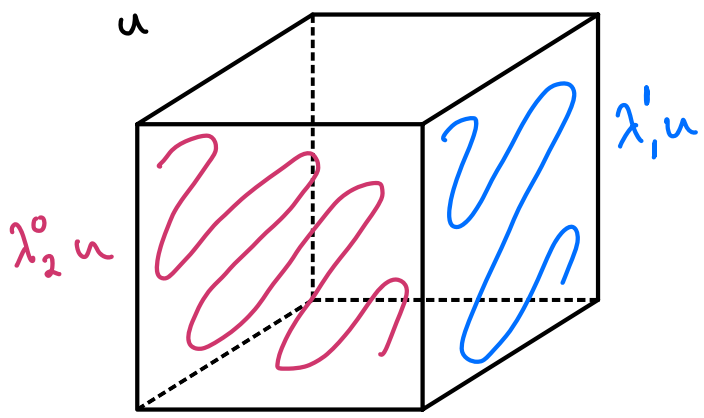
$D_n(X)$ = free abelian group on n-dim degenerate cubes

We can define a border operation by looking at faces of cubes.

Then, we can show that $d^2 = 0$ and $d(D_n(X)) \subseteq D_{n-1}(X)$.

Thus our desired chain complex is

$$C(X) = Q(X)/D(X).$$



$$d(u) = \sum_{i=1}^{\hat{}} (-1)^i (\lambda_i^0 u - \lambda_i^1 u)$$

We can use these cubes to explicitly construct an action of $\pi_1(B)$ on $H(F)$.

Thus, $H(F)$ is a system of local coefficients on B.

For simplicity, we will assume this local system is trivial. That is, $\pi_1(B)$ acts trivially on $H(F)$.

We also use these cubes to explicitly filter $C(E)$.

Define $T^{p,q} \subseteq Q_{p+q}(E)$ as the subgroup generated by the $(p+q)$-dimensional cubes, u, such that the cube $\pi \circ u$ on E does not depend on its last q coordinates.

$$u \in T^{p,q} \iff \pi(u(t_1, \dots, t_p, t_{p+1}, \dots, t_{p+q})) \text{ does } \underline{\text{not}} \text{ depend on } t_{p+1}, \dots, t_{p+q}$$

We filter $C(E)$ by taking the images of $T^{p,q}$ in $C(E)$. It is straightforward to verify this filtration satisfies our earlier requirements.

Using more explicit cube constructions, we can show that

$$E_1^{p,q} = C_p(B) \otimes H_q(F)$$

$$E_2^{p,q} = H_p(B, H_q(F))$$

Speaker Note: Go check out the constructions! It is really remarkable how Serre does this "hands on".

Quick Applications

Our observations so far already suffice to give some neat results!

Euler Characteristic

Let k be a field, e_i, b_i, f_i the dimensions of $H_i(E;k)$, $H_i(B;k)$, $H_i(F;k)$ respectively. Suppose further that B is a CW-complex. If $b_i, f_i < \infty$ for all i and zero for sufficiently large i, then

$$\chi(E) = \chi(B)\chi(F).$$

Proof

For large r we will have $E_r = E_\infty$, so

$$\chi(E) = \chi(E_\infty) = \chi(E_r)$$

However, $\chi(E_{r-1}) = \chi(E_r)$ for all r, so

$$\chi(E_r) = \chi(E_2) = \chi\left(H(B;k) \otimes H(F;k)\right) = \chi(B)\chi(F). \blacksquare$$

Gysin Sequence

Suppose we have a spherical fibration $S^k \to E \to B$ and the local system $H^*(F)$ over B is trivial. Then, we have a long exact sequence $(k \geq 1)$

$$\cdots \to H^i(B) \to H^i(E) \to H^{i-k}(B) \to H^{i+1}(B) \to \cdots .$$

Wang Sequence

Suppose we have a fibration over a sphere $F \to E \to S^k$. Then, we have a long exact sequence $(k \geq 2)$

$$\cdots \leftarrow H_i(E) \leftarrow H_i(F) \leftarrow H_{i-k+1}(F) \leftarrow H_{i+1}(E) \leftarrow \cdots$$

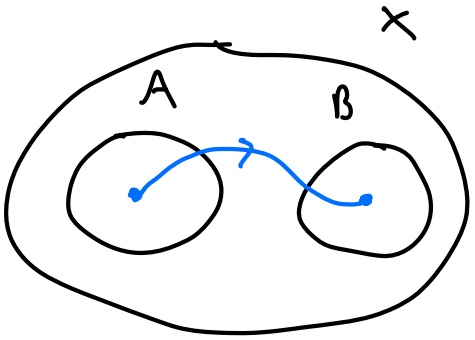
Path Spaces

As foreshadowed earlier, we can write loop spaces as fibers of a certain Serre fibration. Thus, we will be able to apply our technology to compute their homology.

Let X be path-connected, $A, B \subseteq X$. We define

$$E_{A,B} = \left\{ f : I \to X \mid f(0) \in A, f(1) \in B \right\}$$

and give it the compact-open topology.



Furthermore, we define a map

$$p_{A,B} : E_{A,B} \to A \times B$$
$$f \mapsto (f(0), f(1))$$

Proposition

The map $p_{A,B} : E_{A,B} \to A \times B$ is a Serre fibration.

Proof

Suppose we have a commutative square

$$\begin{array}{ccc} W & \xrightarrow{g} & E_{A,B} \\ \downarrow{\scriptstyle \iota_0} & & \downarrow{\scriptstyle p_{A,B}} \\ I \times W & \xrightarrow[(f,f')]{} & A \times B \end{array}$$

The data of the map g is equivalent to the data of a map $G : I \times W \to X$ such that $G(t,x) = g(x)(t)$.

We wish to find $h : I \times W \to E_{A,B}$ satisfying certain conditions.

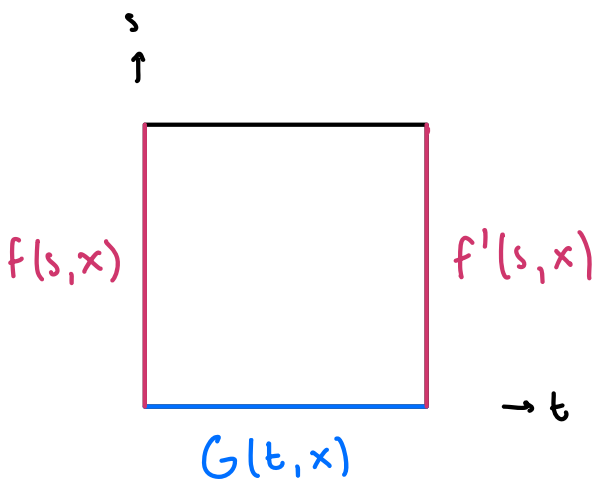
This is the same as finding $H : I \times I \times W \to X$, by setting

$H(t,s,x) = h(s,x)(t)$.

We have a map

$$(I \times \{0\} \times W) \cup (\{0\} \times I \times W) \cup (\{1\} \times I \times W) \to X$$

described in the picture. We can extend this map to $H : I \times I \times W \to X$ and satisfy all



the desired conditions. ∎

Proposition (won't prove this one!)

If A deformation retracts onto a point $x \in X$, the space $E_{A,B}$ has the homotopy type of $A \times E_{x,B}$.

As a corollary it follows that $E_{x,y} \simeq E_{z,t}$ for any points $x, y, z, t \in X$. In particular, we can use any of these to compute the homology of the loop space, Ω.

Our work before tells us that there is a spectral sequence such that

$$E_2^{p,q} = H_p(A \times B, H_q(\Omega)).$$

Observe that the fiber over x of $E_{x,x} \to X$ is exactly Ω, the loop space. Thus, we have a Serre fibration

$$\Omega \longrightarrow E_{x,x} \xrightarrow{p_{x,x}} X.$$

Here's the powerful thing:

Proposition

The space $E_{x,x}$ is contractible.

Proof

Use the map $I \times E_{x,x} \to E_{x,x}$ given by

$$(\theta, f) \mapsto f(\theta \cdot _)$$

∎

Thus, in some sense, the spectral sequence gives the homology of Ω in terms of that of X.

Let's conclude this discussion with my favorite example:

The Homology of ΩS^n

The homology groups of ΩS^n are $(n \geq 2)$

$$H_k(\Omega S^n) = \begin{cases} \mathbb{Z}, & \text{if } (n-1) \mid k \\ 0, & \text{else} \end{cases}$$

Proof

Let $E = E_{x, S^n}$, where $x \in S^n$ is our basepoint. We have a Serre fibration

$$\Omega S^n \to E \to S^n,$$

so we can apply the Wang sequence and get:

$$\cdots \leftarrow H_i(E) \leftarrow H_i(\Omega S^n) \leftarrow H_{i-n+1}(\Omega S^n) \leftarrow H_{i+1}(E) \leftarrow \cdots$$

But, E is contractible, so

$$H_i(\Omega S^n) = H_{i-n+1}(\Omega S^n).$$

The result now follows from $H_0(\Omega S^n) = \mathbb{Z}$ and $H_i(\Omega S^n) = 0$ for $i < 0$. ∎